

Blowup above the exponent one third in viscous Katz–Pavlović dyadic models with small shell ratios

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Abstract

For the viscous Katz–Pavlović dyadic model with shell ratio $\Lambda > 2^{3/2}$, global regularity holds exactly for dissipation exponents $\alpha \geq 1/3$. We prove that this threshold is not universal: for $\Lambda = 6/5, 13/10, 3/2, 17/10$ and $9/5$, suitable nonnegative finitely supported data lose regularity in finite time for every $\alpha \leq \alpha_\Lambda$, where $\alpha_\Lambda > 1/3$; for example $\alpha_{13/10} = 0.378$. The blowup is carried by a self-similar energy front whose amplitude grows by a fixed factor $\kappa > 1$ per shell. The proof reduces the dynamics to one renormalized step of the front and verifies that step in interval arithmetic.

Keywords: dyadic model, Katz–Pavlović model, finite-time blowup, dissipation exponent, self-similar front, computer-assisted proof

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1. Introduction

1.1. The model

Fix a shell ratio $\Lambda > 1$, a viscosity $\nu > 0$ and a dissipation exponent α . The viscous Katz–Pavlović dyadic model is the infinite system

$$\dot{u}_n + \nu \Lambda^{2\alpha n} u_n = \Lambda^{n-1} u_{n-1}^2 - \Lambda^n u_n u_{n+1}, \quad n \geq 0, \quad u_{-1} \equiv 0. \quad (1.1)$$

The coordinate u_n models the velocity at frequency Λ^n ; the quadratic terms move energy from shell $n - 1$ to shell n and conserve $\sum u_n^2$, and the linear term dissipates it. Katz and Pavlović introduced the model for $\Lambda = 2$ as a scalar caricature of the Navier–Stokes nonlinearity [1]; see [2, 3, 4, 5, 6, 7] for related models and [8] for a survey. Writing the shells as $N_n = N_0 \Lambda^n$, as in [9], changes only ν : the substitution $u \mapsto N_0 u$ turns that system into (1.1) with viscosity $\nu N_0^{2\alpha}$.

For $\sigma \geq 0$ let $\|u\|_{H^\sigma}^2 = \sum_{n \geq 0} \Lambda^{2\sigma n} u_n^2$. Following [10, 9], a *finite-energy solution* on an interval $I \ni 0$ is an ℓ^2 -valued map whose coordinates are C^1 on I and satisfy (1.1); it is *smooth* on I if $\sup_{t \in J} \|u(t)\|_{H^\sigma} < \infty$ for every compact $J \subset I$ and every $\sigma > 0$. A nonnegative datum in every H^σ has a unique nonnegative smooth solution on a maximal interval $[0, T_{\max})$ [11, Remark 3.4], [9, Proposition 2.4]; it *loses regularity in finite time* if $T_{\max} < \infty$.

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The known results on regularity are these. Katz and Pavlović proved finite-time blowup for $\alpha < 1/4$ [1]. Cheskidov proved, for every $\Lambda > 1$, that large nonnegative data blow up in $H^{1/3+\gamma}$ when $\alpha < 1/3$, and that all solutions are regular when $\alpha \geq 1/2$ [10, Theorems 4.4 and 5.3]. Barbato, Morandin and Romito proved smoothing in an intermediate range [11]. Recently Zhang proved global regularity of all nonnegative smooth data when

$$\alpha \geq \frac{1}{3} \quad \text{and} \quad 4\alpha - 1 > \frac{\log 2}{2 \log \Lambda} \quad (1.2)$$

[9, Theorem 1.2]. For $\Lambda > 2^{3/2}$ the second condition follows from the first, so $\alpha = 1/3$ is the sharp threshold for these ratios. Zhang notes that the restriction on Λ comes from the method and does not assert that it is optimal. For $\Lambda \leq 2^{3/2}$ the range between $1/3$ and $\min\{1/2, 1/4 + \log 2/(8 \log \Lambda)\}$ was open.

1.2. Main result

The theorem below shows that for small shell ratios the threshold lies strictly above $1/3$. Table 1 lists, for five ratios, the window sizes K_b, K_a of the computation, the exponent α_Λ reached, an enclosure $[G_{lo}, G_{hi}]$ of the growth factor of the front, and two constants γ_Λ and C_Λ . For each row, $w^* = (w_k^*)_{-K_b \leq k \leq K_a}$ is a profile with $w_0^* = 1$, $w_1^* = 1/2$, all $w_k^* \geq 0$; its values are stored to 17 digits in the file `cap/cases/center_L*.txt` of the archive [12], and Table 2 shows part of the profile for $\Lambda = 13/10$. Put

$$h_m^0 = \Lambda^{-m/3} w_{m-K_b}^* \quad (0 \leq m \leq K_b + K_a), \quad h_m^0 = 0 \quad (m > K_b + K_a). \quad (1.3)$$

Table 1: The five verified cases. α_Λ is the largest exponent covered by the computation, $[G_{lo}, G_{hi}]$ encloses the growth factor of one step, $\gamma_\Lambda = 1/3 - \log G_{lo}/\log \Lambda$ and C_Λ are rounded up. The last column is $1/3 + \log G_{lo}/(2 \log \Lambda)$, the largest exponent for which this G_{lo} would close the argument.

Λ	K_b	K_a	α_Λ	$[G_{lo}, G_{hi}]$	γ_Λ	C_Λ	$\frac{1}{3} + \frac{\log G_{lo}}{2 \log \Lambda}$
6/5	80	7	0.377	[1.01628489, 1.01628532]	0.24474	$2.83 \cdot 10^{-4}$	0.37763
13/10	60	6	0.378	[1.02401568, 1.02401602]	0.24288	$8.72 \cdot 10^{-5}$	0.37856
3/2	50	8	0.365	[1.02642827, 1.02642851]	0.26900	$2.82 \cdot 10^{-6}$	0.36550
17/10	40	8	0.348	[1.01599737, 1.01599757]	0.30343	$1.04 \cdot 10^{-6}$	0.34828
9/5	40	8	0.339	[1.00747733, 1.00747753]	0.32066	$2.02 \cdot 10^{-7}$	0.33967

Theorem 1.1. *Let $(\Lambda, K_b, K_a, \alpha_\Lambda)$ be a row of Table 1 and let h^0 be the datum (1.3). Let $\nu > 0$, $0 < \alpha \leq \alpha_\Lambda$, and*

$$A \geq A_0(\nu, \alpha) := 10^{12} \nu \Lambda^{1/3} \Lambda^{(2\alpha-2/3)K_b}. \quad (1.4)$$

- (a) *Let u be a nonnegative finite-energy solution of (1.1) on $[0, T)$ with $u(0) = Ah^0$ and $T > C_\Lambda/A$. There are times $0 = t_{K_b} < t_{K_b+1} < \dots$ with $t_n < C_\Lambda/A$ such that*

$$u_n(t_n) \geq A \Lambda^{-n/3} G_{lo}^{n-K_b} \quad (n \geq K_b).$$

Consequently $\|u(t_n)\|_{H^\sigma} \rightarrow \infty$ for every $\sigma > \gamma_\Lambda$, and u is not smooth on $[0, C_\Lambda/A]$.

- (b) *The smooth solution with datum Ah^0 has $T_{\max} \leq C_\Lambda/A$. More precisely, $T_{\max} = \lim_n t_n$.*

Nonnegative finite-energy solutions with any nonnegative ℓ^2 datum exist for all time [10, Theorem 4.1], so part (a) says that every global such solution with datum Ah^0 loses H^σ regularity, for $\sigma > \gamma_\Lambda$, before the time C_Λ/A . Since $\alpha_\Lambda > 1/3$ in every row, we obtain:

Corollary 1.2. *For $\Lambda \in \{6/5, 13/10, 3/2, 17/10, 9/5\}$ and every $\alpha \in [1/3, \alpha_\Lambda]$, global regularity of (1.1) fails for some nonnegative finitely supported datum. In particular, the condition $\alpha \geq 1/3$ alone does not imply global regularity, and the restriction on Λ in (1.2) cannot be dropped.*

For $\Lambda = 13/10$ the second condition in (1.2) requires $\alpha > 0.5802$, so the corollary is consistent with [9]; the range $0.378 < \alpha < 1/2$ remains open for this ratio (Figure 2(b)).

Remark 1.3. The proof uses only that the front coordinates of the datum at shell K_b satisfy (S1)–(S5) of Definition 2.5. Hence the conclusions hold for every datum Ag with $g \geq 0$, $g_m = \Lambda^{-m/3}v_{m-K_b}$ for $0 \leq m \leq K_b + K_a$, where $v_0 = 1$, $v_1 = 1/2$ and the other v_k lie in the box B of (S2), and with $g_m \leq \Lambda^{-m/3}\theta_{m-K_b}$ for $m > K_b + K_a$. These data form a relatively open subset of a hyperplane; we have not checked the step for states near the section but off it. The time C_Λ/A scales like the inverse amplitude; by the scaling $u \mapsto au(a \cdot)$, which maps viscosity ν to $a\nu$, the theorem equivalently gives blowup of the fixed datum h^0 for every viscosity $\nu \leq 10^{-12}\Lambda^{-1/3}\Lambda^{(2/3-2\alpha)K_b}$.

1.3. The mechanism

Set $y_n = \Lambda^{n/3}u_n$. Kolmogorov scaling corresponds to y_n of order one, and in these variables the inviscid model has the rate ratio $r = \Lambda^{2/3}$ between neighbouring shells (Section 2). Energy injected at low shells forms a front that moves to higher shells. After the front has passed shell n , the profile seen from the front repeats itself up to a factor: the amplitude of the leading shell grows by a factor $\kappa(\Lambda)$ per shell, and the passage through shell n takes a time proportional to $(r\kappa)^{-n}$. Viscosity at the front is measured by $\varepsilon_n \propto \Lambda^{2\alpha n}/(r\kappa)^n$, which tends to zero when

$$\Lambda^{2\alpha} < r\kappa \quad \Longleftrightarrow \quad \alpha < \frac{1}{3} + \frac{\log \kappa}{2 \log \Lambda}. \quad (1.5)$$

If $\kappa = 1$, as for a front with Kolmogorov scaling, this is Cheskidov's range $\alpha < 1/3$. Numerically, $\kappa(\Lambda) > 1$ exactly when $\Lambda < \Lambda_c \approx 1.8754$ (Section 6 and Figure 2(a)). Then y_n at the front grows like κ^n , u_n decays only like $\Lambda^{-\gamma n}$ with $\gamma = 1/3 - \log \kappa / \log \Lambda < 1/3$, and the window (1.5) extends beyond $1/3$. Figure 1 shows the front for $\Lambda = 13/10$.

1.4. Method and what the computer checks

We write the solution in coordinates attached to the front (Section 2) and show that one renormalized step maps a box B of states into itself, with growth factor at least $G_{10} > 1$ and with the viscosity parameter decreasing (Proposition 3.1). Iterating the step proves Theorem 1.1 (Section 3).

The following parts are proved by hand: the reduction to front coordinates (Lemmas 2.2 and 2.4), the iteration that gives Theorem 1.1 from Proposition 3.1, and the five lemmas of Section 4 that control the infinitely many shells outside a finite window, the error of truncating to that window, and the crossing of the section. Eight of these elementary lemmas are also formalized in Lean 4 (Appendix Appendix A).

What remains is a finite list of inequalities, (V1)–(V9) in Section 5, about one system of $K_b + K_a + 1$ ordinary differential equations with parameters $\varepsilon \in [0, 10^{-12}]$ and $\alpha \in [0, \alpha_\Lambda]$.

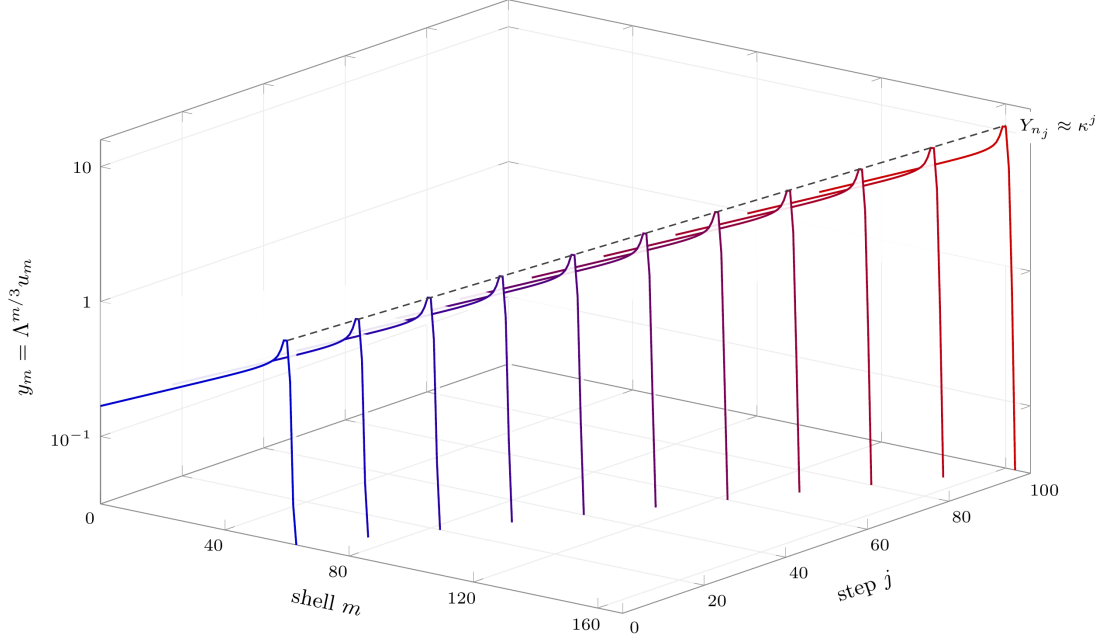


Figure 1: The front for $\Lambda = 13/10$, $\alpha = 3/8$: the compensated amplitudes $y_m = \Lambda^{m/3} u_m$ at the start of every tenth renormalized step j , computed from the datum h^0 (Section 6). Each step moves the front by one shell and multiplies its height by $\kappa \approx 1.024016$; the dashed line has slope $\log_{10} \kappa$ per step. Values below $10^{-1.5}$ are cut off.

These are verified by a computer in interval arithmetic with the CAPD library [13, 14]. The computation encloses the flow of the truncated system over one step, together with its derivative, for all states in B at once. Thus the proof is computer-assisted in the usual sense [15]: it is rigorous provided the CAPD library and our program, both public, are correct. Each case takes a few seconds (Table 3).

1.5. Scope

The result concerns the dyadic model, not the Navier–Stokes equations. It shows that the threshold $1/3$, which matches the scaling of the three-dimensional equations and is sharp for $\Lambda > 2^{3/2}$, is an artifact of the shell geometry for small Λ : there the front develops anomalous scaling, and the dissipation needed to stop it is larger. Section 6 records the numerical picture for all Λ and a conjecture on the threshold; Section 7 lists open questions.

2. Front coordinates

Throughout, u is a nonnegative finite-energy solution of (1.1) on $[0, T)$, and $r = \Lambda^{2/3}$.

Lemma 2.1 (Positivity). *If $u(0) \geq 0$ coordinatewise, then $u_n(t) \geq 0$ for all n and t .*

Proof. Put $a_n(t) = \nu \Lambda^{2\alpha n} + \Lambda^n u_{n+1}(t)$, a continuous function. Then

$$\frac{d}{dt} \left(u_n e^{\int_0^t a_n} \right) = \Lambda^{n-1} u_{n-1}^2 e^{\int_0^t a_n} \geq 0,$$

so $u_n(t) \geq u_n(0) e^{-\int_0^t a_n} \geq 0$. □

Lemma 2.2 (Scaling). *Let $y_n(\tau) = \Lambda^{n/3} u_n(\Lambda^{1/3} \tau)$ and $\tilde{\nu} = \Lambda^{1/3} \nu$. Then*

$$\frac{dy_n}{d\tau} = r^n (y_{n-1}^2 - y_n y_{n+1}) - \tilde{\nu} \Lambda^{2\alpha n} y_n, \quad y_{-1} \equiv 0. \quad (2.1)$$

Proof. With $t = \Lambda^{1/3} \tau$ and $u_m = \Lambda^{-m/3} y_m$,

$$\frac{dy_n}{d\tau} = \Lambda^{1/3} \Lambda^{n/3} \left(\Lambda^{n-1} \Lambda^{-2(n-1)/3} y_{n-1}^2 - \Lambda^n \Lambda^{-n/3-(n+1)/3} y_n y_{n+1} - \nu \Lambda^{2\alpha n} \Lambda^{-n/3} y_n \right),$$

and both quadratic coefficients equal $\Lambda^{2n/3} = r^n$. $\square$

Definition 2.3 (Front coordinates). Let $n \geq 0$ and $\tau_n \in [0, \Lambda^{-1/3} T)$ with $Y_n := y_n(\tau_n) > 0$. The *front coordinates* at (n, τ_n) are

$$Y_k(s) = \frac{y_{n+k}(\tau_n + s/(r^n Y_n))}{Y_n} \quad (k \geq -n),$$

the *time of the step* is $s \geq 0$, and the *viscosity parameter* is $\varepsilon_n = \tilde{\nu} \Lambda^{2\alpha n} / (r^n Y_n)$.

The subscript of Y_n refers to the shell and that of Y_k to the position relative to the front; no confusion will arise. Put $R_k = r^k$ and $E_k = \varepsilon_n \Lambda^{2\alpha k}$. Differentiating, and using (2.1) with $y_{n+k} = Y_n Y_k$,

$$\frac{dY_k}{ds} = R_k (Y_{k-1}^2 - Y_k Y_{k+1}) - E_k Y_k \quad (k \geq -n), \quad Y_{-n-1} \equiv 0. \quad (2.2)$$

Indeed $\frac{dY_k}{ds} = \frac{1}{r^n Y_n^2} \frac{dy_{n+k}}{d\tau}$, and $\frac{1}{r^n Y_n^2} (r^{n+k} Y_n^2 (Y_{k-1}^2 - Y_k Y_{k+1}) - \tilde{\nu} \Lambda^{2\alpha(n+k)} Y_n Y_k)$ is the right side of (2.2). Fix $c = 1/2$ and put $h(Y) = Y_2 - cY_1$.

Lemma 2.4 (Renormalization). *Suppose that $Y_0(0) = 1$, $Y_1(0) = c$, and that at some $s_* > 0$ we have $h(Y(s_*)) = 0$ and $G := Y_1(s_*) > 0$. Put $\tau_{n+1} = \tau_n + s_*/(r^n Y_n)$. Then $y_{n+1}(\tau_{n+1}) = GY_n$, the front coordinates Y' at $(n+1, \tau_{n+1})$ satisfy*

$$Y'_k(0) = \frac{Y_{k+1}(s_*)}{G} \quad (k \geq -n-1), \quad Y'_0(0) = 1, \quad Y'_1(0) = c,$$

and $\varepsilon_{n+1} = \varepsilon_n \Lambda^{2\alpha} / (rG)$.

Proof. By definition $y_{n+1}(\tau_{n+1}) = Y_n Y_1(s_*) = GY_n$, and $Y'_k(0) = y_{n+1+k}(\tau_{n+1}) / (GY_n) = Y_{k+1}(s_*) / G$. In particular $Y'_0(0) = 1$ and $Y'_1(0) = Y_2(s_*) / G = cY_1(s_*) / G = c$. Finally $\varepsilon_{n+1} = \tilde{\nu} \Lambda^{2\alpha(n+1)} / (r^{n+1} GY_n) = \varepsilon_n \Lambda^{2\alpha} / (rG)$. $\square$

We call $Y_{-n}, \dots, Y_{-K_b-1}$ the *deep tail*, $Y_{-K_b}, \dots, Y_{K_a}$ the *window* and $Y_{K_a+1}, Y_{K_a+2}, \dots$ the *ahead tail*. The deep tail is empty when $n = K_b$.

Definition 2.5 (Admissible states). Fix a row of Table 1 with profile w^* and the radii $\rho_k > 0$ stored in `cap/cases/radii_L*.txt` [12]. Let $W_d = 1$, $\theta = 10^{-31}$, $\theta_k = \theta 2^{-(k-K_a-1)}$ for $k > K_a$, and $\varepsilon_0 = 10^{-12}$. For $n \geq K_b$, the set $\mathcal{S}_n$ consists of the pairs (Y, ε) , $Y = (Y_k)_{k \geq -n}$, with

(S1) $Y_k \geq 0$ for all k , $Y_0 = 1$ and $Y_1 = c$;

(S2) $|Y_k - w_k^*| < \rho_k$ for $-K_b \leq k \leq K_a$, $k \notin \{0, 1\}$;

(S3) $Y_k \leq W_d$ for $-n \leq k < -K_b$;

(S4) $Y_k \leq \theta_k$ for $k > K_a$;

(S5) $0 \leq \varepsilon \leq \varepsilon_0$.

We write B for the open box in (S2).

3. The step proposition and the proof of the theorem

Proposition 3.1 (One step). *Fix a row of Table 1, let $0 \leq \alpha \leq \alpha_\Lambda$, $n \geq K_b$, and let s_{end} be the number in Table 3. Let Y be a nonnegative solution of (2.2), with coordinates of class C^1 , on an interval $[0, s_1]$, with $(Y(0), \varepsilon_n) \in \mathcal{S}_n$.*

(i) *If $s_1 \geq s_{\text{end}}$, then $h(Y(s))$ has exactly one zero s_* in $[0, s_{\text{end}}]$; it is negative on $[0, s_*)$ and positive on $(s_*, s_{\text{end}}]$.*

(ii) *In that case $G = Y_1(s_*) \in [G_{\text{lo}}, G_{\text{hi}}]$, and the state*

$$\left((Y_{k+1}(s_*)/G)_{k \geq -n-1}, \varepsilon_n \Lambda^{2\alpha} / (rG) \right)$$

lies in $\mathcal{S}_{n+1}$.

(iii) *For every $s_1 \leq s_{\text{end}}$, the coordinates satisfy on $[0, s_1]$*

$$Y_k \leq \frac{W_d}{1 - W_d R_{-K_b-1} s_{\text{end}}} \quad (k < -K_b), \quad Y_k \leq M \quad (|k| \leq K_b + K_a),$$

$$Y_k \leq 2\theta_k \quad (k > K_a),$$

with a constant M depending only on the row.

The proof of Proposition 3.1 occupies Sections 4 and 5. We now derive Theorem 1.1.

Proof of Theorem 1.1. (a) Let y, τ be as in Lemma 2.2, and $T' = \Lambda^{-1/3}T$. Put $n_0 = K_b$, $\tau_{n_0} = 0$, $Y_{n_0} = y_{K_b}(0) = A$. The front coordinates at $(n_0, 0)$ are $Y_k(0) = w_k^*$ for $-K_b \leq k \leq K_a$ and $Y_k(0) = 0$ for $k > K_a$, and the deep tail is empty. By (1.4),

$$\varepsilon_{n_0} = \frac{\tilde{\nu} \Lambda^{2\alpha K_b}}{r^{K_b} A} = \frac{\nu \Lambda^{1/3} \Lambda^{(2\alpha-2/3)K_b}}{A} \leq 10^{-12},$$

so $(Y(0), \varepsilon_{n_0}) \in \mathcal{S}_{n_0}$ (the profile has $w^* \geq 0$ and lies in B).

Let $D_\Lambda = s_{\text{end}} / (r^{n_0} A) \sum_{i \geq 0} (rG_{\text{lo}})^{-i}$. We show by induction on $n \geq n_0$: there are times $\tau_{n_0} < \dots < \tau_n$ with

$$\tau_n + \frac{s_{\text{end}}}{r^n Y_n} \leq \sum_{j=n_0}^n \frac{s_{\text{end}}}{r^j Y_j} \leq D_\Lambda, \quad Y_j \geq A G_{\text{lo}}^{j-n_0}, \quad (3.1)$$

where $Y_j = y_j(\tau_j)$, and the front coordinates at (n, τ_n) , together with ε_n , lie in $\mathcal{S}_n$. For $n = n_0$ this was shown. Assume it for n . Since $\Lambda^{1/3} D_\Lambda = C_\Lambda / A < T$ (the value of C_Λ in Table 1 is $\Lambda^{1/3} s_{\text{end}} r^{-K_b} (1 - (rG_{\text{lo}})^{-1})^{-1}$ rounded up), the solution exists on $[\tau_n, \tau_n + s_{\text{end}} / (r^n Y_n)]$, so

the front coordinates solve (2.2) on $[0, s_{\text{end}}]$. Proposition 3.1 gives $s_* \leq s_{\text{end}}$ and $G \geq G_{\text{lo}}$, and Lemma 2.4 gives $\tau_{n+1} = \tau_n + s_*/(r^n Y_n)$, $Y_{n+1} = GY_n \geq AG_{\text{lo}}^{n+1-n_0}$ and an admissible state at $n+1$. Then $\tau_{n+1} + s_{\text{end}}/(r^{n+1}Y_{n+1}) \leq \sum_{j=n_0}^{n+1} s_{\text{end}}/(r^j Y_j)$, and the sum is at most $s_{\text{end}} r^{-n_0} A^{-1} \sum_{i \geq 0} (rG_{\text{lo}})^{-i} = D_\Lambda$.

Put $t_n = \Lambda^{1/3} \tau_n < C_\Lambda/A$. Then $u_n(t_n) = \Lambda^{-n/3} Y_n \geq A \Lambda^{-n/3} G_{\text{lo}}^{n-K_b}$ and

$$\|u(t_n)\|_{H^\sigma} \geq \Lambda^{\sigma n} u_n(t_n) \geq A G_{\text{lo}}^{-K_b} \left(\Lambda^{\sigma-1/3} G_{\text{lo}} \right)^n,$$

which tends to infinity when $\Lambda^{\sigma-1/3} G_{\text{lo}} > 1$, that is, $\sigma > 1/3 - \log G_{\text{lo}} / \log \Lambda$; Table 1 lists this number rounded up. Since all t_n lie in $[0, C_\Lambda/A]$, u is not smooth there.

(b) The smooth solution is a nonnegative finite-energy solution on $[0, T_{\text{max}})$. If $T_{\text{max}} > C_\Lambda/A$, it would be smooth on $[0, C_\Lambda/A]$, contradicting (a). Hence $T_{\text{max}} \leq C_\Lambda/A < \infty$.

Now run the induction of (a) for this solution, with $T = T_{\text{max}}$; step n can be carried out if $t_n + \Lambda^{1/3} s_{\text{end}}/(r^n Y_n) < T_{\text{max}}$. Suppose that this fails at some n . For every $t \in [t_n, T_{\text{max}})$ the front coordinates at (n, τ_n) solve (2.2) on an interval $[0, s_1]$ with $s_1 < s_{\text{end}}$ that reaches the time t , and the state at $s = 0$ is admissible. By Proposition 3.1(iii), with $M' = \max\{M, W_d/(1 - W_d R_{-K_b-1} s_{\text{end}})\}$, we get $y_{n+k}(\tau) \leq M' Y_n$ for $k \leq K_a$ and $y_{n+k}(\tau) \leq 2\theta_k Y_n$ for $k > K_a$, at the corresponding τ . With $\Lambda^m u_m = \Lambda^{2m/3} y_m$ and $\theta_k = 2\theta 2^{-(k-K_a)}$ this gives

$$\sup_m \Lambda^m u_m(t) \leq Y_n \Lambda^{2(n+K_a)/3} \max\left\{M', 4\theta \sup_{j \geq 1} \left(\Lambda^{2/3}/2\right)^j\right\} \quad (t_n \leq t < T_{\text{max}}),$$

which is finite because $\Lambda^{2/3} < 2$ for all five ratios. On the compact interval $[0, t_n]$ the solution is smooth, so $\sup_m \Lambda^m u_m \leq \|u\|_{H^1}$ is bounded there too. This contradicts the continuation criterion of [9, Proposition 2.4]: if $T_{\text{max}} < \infty$, then $\limsup_{t \uparrow T_{\text{max}}} \sup_m \Lambda^q u_m(t) = \infty$ for every $q > \max\{0, 1 - 2\alpha\}$, in particular for $q = 1$.

Hence every step is carried out, all $t_n < T_{\text{max}}$, and $t_* := \lim_n t_n \leq T_{\text{max}}$. If $t_* < T_{\text{max}}$, the solution would be smooth on the compact interval $[0, t_*]$, which contains all t_n , whereas $\|u(t_n)\|_{H^\sigma} \rightarrow \infty$ by (a). Hence $T_{\text{max}} = t_*$. $\square$

Remark 3.2. The proof of part (a) uses neither the energy inequality nor the local theory; part (b) uses the local theory only through the existence of the smooth solution and, for the identity $T_{\text{max}} = \lim t_n$, through the continuation criterion.

4. Lemmas for one step

Fix a row of Table 1, $\alpha \in [0, \alpha_\Lambda]$, $n \geq K_b$, and a solution Y of (2.2) on $[0, s_1]$ as in Proposition 3.1, with $s_1 \leq s_{\text{end}}$. The window $Z = (Y_k)_{-K_b \leq k \leq K_a} \in \mathbb{R}^W$, $W = \{-K_b, \dots, K_a\}$, satisfies

$$Z' = F(Z) + e(s), \tag{4.1}$$

where F is the right side of (2.2) on the window with Y_{-K_b-1} and Y_{K_a+1} replaced by 0 (the *truncated system*), and the *truncation error* e has only two nonzero coordinates,

$$e_{-K_b} = R_{-K_b} Y_{-K_b-1}^2 \geq 0, \quad e_{K_a} = -R_{K_a} Y_{K_a} Y_{K_a+1} \leq 0 \tag{4.2}$$

($e_{-K_b} = 0$ if $n = K_b$). Since $K_a \geq 3$, the coordinates e_1, e_2 vanish, so $h = Z_2 - cZ_1$ has the same derivative $F_2 - cF_1$ along (4.1) and along the truncated system. All coordinates are nonnegative by Lemma 2.1, which applies to (2.2) by the same proof. We write $\tilde{Z}$ for the solution of the truncated system $\tilde{Z}' = F(\tilde{Z})$ with $\tilde{Z}(0) = Z(0)$.

Lemma 4.1 (Deep tail). *Let $\bar{R} = R_{-K_b-1}$ and suppose $W_d \bar{R} s_{\text{end}} < 1$ and $Y_k(0) \leq W_d$ for $k < -K_b$. Then $Y_k(s) \leq \widehat{W}(s) := W_d/(1 - W_d \bar{R} s)$ for $-n \leq k < -K_b$ and $s \in [0, s_1]$.*

Proof. For $k < -K_b$ we have $R_k \leq \bar{R}$ and, dropping nonpositive terms, $Y'_k \leq \bar{R} Y_{k-1}^2$, where Y_{k-1} is a deep-tail coordinate or $Y_{-n-1} = 0$. For $\delta > 0$ let $B_\delta(s) = (W_d + \delta)/(1 - (W_d + \delta)(\bar{R} + \delta)s)$ on the interval where the denominator is positive; $B'_\delta = (\bar{R} + \delta)B_\delta^2$. If some Y_k reaches B_δ , let s_2 be the first time at which $\max_{k < -K_b} (Y_k - B_\delta) = 0$ (a maximum of finitely many continuous functions, negative at $s = 0$), and k an index attaining it. Then $(Y_k - B_\delta)'(s_2) \geq 0$, while $Y'_k(s_2) \leq \bar{R} Y_{k-1}(s_2)^2 \leq \bar{R} B_\delta(s_2)^2 < B'_\delta(s_2)$. Hence $Y_k < B_\delta$ on the common interval, and letting $\delta \rightarrow 0$ proves the claim. $\square$

Lemma 4.2 (Ahead tail). *Let $q_{K_a} \geq \sup_{[0, s_1]} Y_{K_a}$, put $q_{K_a+1} = \theta + R_{K_a+1} s_{\text{end}} q_{K_a}^2$ and $q_k = \theta_k + R_k s_{\text{end}} q_{k-1}^2$ for $k \geq K_a + 2$, and suppose $Y_k(0) \leq \theta_k$ for $k > K_a$.*

- (a) $Y_k(s) \leq q_k$ for all $k > K_a$ and $s \in [0, s_1]$.
- (b) Suppose (A1) $R_{K_a+1} s_{\text{end}} q_{K_a}^2 \leq \theta$, (A2) $r \leq 2$, (A3) $c_1 := 4R_{K_a+2} s_{\text{end}} \theta \leq 1/2$, and (A4) $1/2 + c_1 \leq G_*$ for some $G_* > 0$. Then $q_k \leq 2\theta_k$ for all $k > K_a$, and $q_k \leq G_* \theta_{k-1}$ for all $k \geq K_a + 2$.

Proof. (a) By induction on k : $Y'_k \leq R_k Y_{k-1}^2$, so $Y_k(s) \leq Y_k(0) + R_k \int_0^s Y_{k-1}^2 \leq \theta_k + R_k s_{\text{end}} q_{k-1}^2$.

(b) Write $a_k = 4R_k s_{\text{end}} \theta_{k-1}$ for $k \geq K_a + 2$; then $a_{K_a+2} = c_1$ and $a_{k+1}/a_k = r/2 \leq 1$, so $a_k \leq c_1 \leq 1/2$. First, $q_{K_a+1} \leq 2\theta = 2\theta_{K_a+1}$ by (A1). If $q_{k-1} \leq 2\theta_{k-1}$, then, as $\theta_k = \theta_{k-1}/2$,

$$q_k \leq \theta_k + 4R_k s_{\text{end}} \theta_{k-1}^2 = \theta_{k-1} \left(\frac{1}{2} + a_k \right),$$

which is at most $\theta_{k-1} = 2\theta_k$ because $a_k \leq 1/2$, and at most $G_* \theta_{k-1}$ because $1/2 + a_k \leq 1/2 + c_1 \leq G_*$. $\square$

Lemma 4.3 (Truncation error). *Let $H \subset \mathbb{R}^W$ be a box containing $\tilde{Z}(s)$ for $s \in [0, s_1]$, let $D \in \mathbb{R}^W$ have positive entries, and let M be a matrix with*

$$M_{ii} \geq \sup \partial_i F_i, \quad M_{ij} \geq \sup |\partial_j F_i| \quad (i \neq j),$$

the suprema taken over $H + [-D, D]$ and over the parameters. Let $\zeta \in \mathbb{R}^W$ have positive entries and $\lambda > 0$ satisfy $M\zeta \leq \lambda\zeta$. Let $\bar{e} \in \mathbb{R}^W$ satisfy $|e(s)| \leq \bar{e}$ at every $s \in [0, s_1]$ such that $|Z - \tilde{Z}| \leq D$ on $[0, s]$, and let $\eta \geq 0$ with $\bar{e} \leq \eta\zeta$. If $z := \eta\zeta (e^{\lambda s_{\text{end}}} - 1)/\lambda < D$, then

$$|Z(s) - \tilde{Z}(s)| \leq \eta\zeta \frac{e^{\lambda s} - 1}{\lambda} \leq z \quad (0 \leq s \leq s_1).$$

All inequalities between vectors are coordinatewise.

This is the usual comparison argument for quasimonotone systems [16]; we include the proof because the error bound $\bar{e}$ is only available while $|Z - \tilde{Z}| \leq D$.

Proof. Let $d = Z - \tilde{Z}$ and $s_2 = \sup\{s \in [0, s_1] : |d| \leq D \text{ on } [0, s]\}$; $s_2 > 0$ because $d(0) = 0$. On $[0, s_2]$ both Z and $\tilde{Z}$ lie in the convex box $H + [-D, D]$, so $d' = A(s)d + e(s)$ with $A(s) = \int_0^1 DF(\tilde{Z} + \vartheta d) d\vartheta$, $A_{ii} \leq M_{ii}$, $|A_{ij}| \leq M_{ij}$ for $i \neq j$, and $|e| \leq \bar{e}$.

For $\delta > 0$ put $w^\delta(s) = \eta\zeta(e^{\lambda s} - 1)/\lambda + \delta e^{(\lambda+1)s}\zeta$. We claim $|d_i| < w_i^\delta$ on $[0, s_2]$ for all i . This holds at $s = 0$. Otherwise let s_3 be the first time with $|d_i(s_3)| = w_i^\delta(s_3)$ for some i .

Then $d_i(s_3) \neq 0$, the function $|d_i| - w_i^\delta$ is differentiable at s_3 , negative before and zero at s_3 , so its derivative there is ≥ 0 . But

$$|d_i|' = \text{sgn}(d_i) d_i' \leq A_{ii}|d_i| + \sum_{j \neq i} |A_{ij}| |d_j| + |e_i| \leq (Mw^\delta)_i + \bar{e}_i,$$

using $|d_j| \leq w_j^\delta$ and $M_{ij} \geq 0$ for $j \neq i$. Since $M\zeta \leq \lambda\zeta$ and $\bar{e} \leq \eta\zeta$,

$$(Mw^\delta)_i + \bar{e}_i \leq \eta(e^{\lambda s} - 1)\zeta_i + \delta\lambda e^{(\lambda+1)s}\zeta_i + \eta\zeta_i < \eta e^{\lambda s}\zeta_i + \delta(\lambda+1)e^{(\lambda+1)s}\zeta_i = (w_i^\delta)'(s)$$

at $s = s_3$, a contradiction. Letting $\delta \rightarrow 0$ gives $|d| \leq w^0 \leq z$ on $[0, s_2]$. As $z < D$, continuity shows that $s_2 < s_1$ is impossible. $\square$

Lemma 4.4 (Section crossing). *Let $0 = \sigma_0 < \sigma_1 < \dots < \sigma_J = s_{\text{end}}$ and let E_j be boxes with $\tilde{Z}(s) \in E_j$ for $s \in [\sigma_j, \sigma_{j+1}]$, for every initial state in B . Let $z \geq 0$ bound $|Z - \tilde{Z}|$ on $[0, s_{\text{end}}]$ and put $E_j^z = E_j + [-z, z]$, $\xi = z_2 + cz_1$. Suppose that $h < 0$ on $\bar{B}$, that for every j either $h < 0$ on E_j^z , or $\nabla h \cdot F \geq m$ on E_j^z for a constant $m > 0$, and that $h > \xi$ on an enclosure E_{end} of $\tilde{Z}(s_{\text{end}})$. Then:*

- (a) $h(\tilde{Z}(\cdot))$ and $h(Z(\cdot))$ each have exactly one zero in $[0, s_{\text{end}}]$, at $\tilde{s}$ and s_* , are negative before it and positive after;
- (b) $|s_* - \tilde{s}| \leq \xi/m$;
- (c) $|Z(s_*) - \tilde{Z}(\tilde{s})| \leq z + \bar{F}\xi/m$, where $\bar{F}$ bounds $|F + e|$ on the boxes E_j^z of the second kind.

Proof. Let X be $\tilde{Z}$ or Z ; then $X(s) \in E_j^z$ for $s \in [\sigma_j, \sigma_{j+1}]$, and $\frac{d}{ds}h(X) = \nabla h \cdot F(X)$ because $e_1 = e_2 = 0$. Whenever $h(X(s)) \geq 0$, the box containing $X(s)$ is of the second kind, so $\frac{d}{ds}h(X) \geq m$. Hence once $h(X) \geq 0$ it increases strictly. Since $h(X(0)) < 0$ (both start at the same point of $\bar{B}$) and $h(X(s_{\text{end}})) > 0$ (for $X = Z$ because $|h(Z) - h(\tilde{Z})| \leq \xi$), there is exactly one zero, which proves (a). On the interval between $\tilde{s}$ and s_* the function $h(Z)$ is ≥ 0 (if $s_* < \tilde{s}$) or $h(\tilde{Z})$ is ≥ 0 (if $\tilde{s} < s_*$), so all boxes met there are of the second kind and $\frac{d}{ds}h(Z) \geq m$ on it. Since $|h(Z(\tilde{s}))| = |h(Z(\tilde{s})) - h(\tilde{Z}(\tilde{s}))| \leq \xi$, the mean value theorem gives (b). For (c) write $Z(s_*) - \tilde{Z}(\tilde{s}) = (Z(s_*) - Z(\tilde{s})) + (Z(\tilde{s}) - \tilde{Z}(\tilde{s}))$ and bound the first term by $\bar{F}|s_* - \tilde{s}|$. $\square$

Lemma 4.5 (Image of the box). *For $P \in \mathbb{R}^W$ with $P_1 > 0$ let $\Phi(P)_k = P_{k+1}/P_1$ for $-K_b \leq k < K_a$ and $\Phi(P)_{K_a} = 0$. Let $\mathcal{P} \subset \mathbb{R}^W$ be a box with $P_1 > 0$ on $\mathcal{P}$, containing $Z(s_*)$ and $\tilde{Z}(\tilde{s})$. Then*

$$\Phi(Z(s_*)) \in \Phi(\tilde{Z}(\tilde{s})) + D\Phi(\mathcal{P})(Z(s_*) - \tilde{Z}(\tilde{s})),$$

where $D\Phi(\mathcal{P})$ is any interval matrix enclosing the Jacobian of Φ on $\mathcal{P}$, and the renormalized window of Lemma 2.4 equals $\Phi(Z(s_*)) + (0, \dots, 0, Y_{K_a+1}(s_*)/G)$.

Proof. The first claim is the mean value theorem applied coordinatewise on the segment from $\tilde{Z}(\tilde{s})$ to $Z(s_*)$, which lies in $\mathcal{P}$. The second is Lemma 2.4: the renormalized window coordinate $k < K_a$ is $Y_{k+1}(s_*)/G = \Phi(Z(s_*))_k$, and coordinate K_a is $Y_{K_a+1}(s_*)/G$. $\square$

5. The computation

5.1. What is computed

For each row of Table 1 the program `cap/verify_step.cpp` [12] works with the truncated system on the window, with the parameters $\varepsilon \in [0, \varepsilon_0]$ and $\alpha \in [0, \alpha_\Lambda]$ entering as intervals, and with the box $\bar{B}$ (the closure of B , with $Z_0 = 1$, $Z_1 = c$). All numbers are intervals with outward rounding. It computes:

- (C1) an enclosure $\mathcal{P}_0$ of the first crossing point $\tilde{Z}(\tilde{s})$ of the section $\{h = 0\}$ from $\{h < 0\}$, for all initial states in $\bar{B}$, with an enclosure of $\tilde{s}$, and an enclosure of the derivative of this Poincaré map; from the derivative and the image of the centre it forms the mean value enclosure and intersects it with $\mathcal{P}_0$;
- (C2) boxes E_j enclosing $\tilde{Z}$ on time intervals covering $[0, s_{\text{end}}]$, with $s_{\text{end}} = \sup \tilde{s} + 0.02$ (0.05 for $\Lambda = 3/2$), and an enclosure E_{end} of $\tilde{Z}(s_{\text{end}})$; H is the hull of the E_j and of $\bar{B}$.

These use the C^1 Lohner algorithm of CAPD with Taylor order 20 [13, 14]. It then checks:

- (V1) $W_d R_{-K_b-1} s_{\text{end}} < 1$; put $\widehat{W} = W_d / (1 - W_d R_{-K_b-1} s_{\text{end}})$.
- (V2) The hypotheses of Lemma 4.3 with $\bar{e}_{-K_b} = R_{-K_b} \widehat{W}^2$, $\bar{e}_{K_a} = R_{K_a} q_{K_a} q_{K_a+1}$, $\bar{e}_k = 0$ otherwise, where q_{K_a} bounds Z_{K_a} on $H + [-D, D]$ and q_{K_a+1} is as in Lemma 4.2; the vector D is found by iterating $D \leftarrow 2z$, and the check is the strict inequality $z < D$.
- (V3) The hypotheses of Lemma 4.4: $h < 0$ on $\bar{B}$, the sign or slope condition on every E_j^z , and $h > \xi$ on E_{end} .
- (V4) With $\mathcal{P} = \mathcal{P}_0 + [-1, 1](z + \bar{F}\xi/m)$ and $G_{\text{lo}} = \inf_{\mathcal{P}} P_1$: conditions (A1)–(A4) of Lemma 4.2 with $G_* = G_{\text{lo}}$.
- (V5) $G_{\text{lo}} > 1$ and $G_{\text{hi}} = \sup_{\mathcal{P}} P_1$ (the values of Table 1).
- (V6) For every $k \in W \setminus \{0, 1\}$, the enclosure of the renormalized coordinate k given by Lemma 4.5, with $[0, q_{K_a+1}/G_{\text{lo}}]$ added at $k = K_a$, lies in $(w_k^* - \rho_k, w_k^* + \rho_k)$.
- (V7) $\sup_{\mathcal{P}} P_{-K_b}/G_{\text{lo}} \leq W_d$ and $\widehat{W}/G_{\text{lo}} \leq W_d$.
- (V8) $\Lambda^{2\alpha_\Lambda}/(rG_{\text{lo}}) < 1$.
- (V9) The mean value enclosure in (C1) and its intersection are nonempty (a consistency check).

Proof of Proposition 3.1, given (V1)–(V9). Lemma 4.1 applies by (V1) and (S3), and gives $e_{-K_b} \leq \bar{e}_{-K_b}$. While $|Z - \tilde{Z}| \leq D$, the window coordinate Z_{K_a} is at most q_{K_a} , so Lemma 4.2(a) gives $Y_{K_a+1} \leq q_{K_a+1}$ and $|e_{K_a}| \leq \bar{e}_{K_a}$. Lemma 4.3 with (V2) gives $|Z - \tilde{Z}| \leq z$ on $[0, s_1]$, hence Z stays in $H + [-z, z]$ and q_{K_a} bounds Z_{K_a} on all of $[0, s_1]$; with Lemma 4.2(b) under (V4) this proves (iii), with $M = \sup(H + [-z, z])$. Now let $s_1 \geq s_{\text{end}}$. Lemma 4.4 with (V3) gives (i) and $Z(s_*) \in \mathcal{P}$, so $G = Z_1(s_*) \in [G_{\text{lo}}, G_{\text{hi}}]$. The new state satisfies (S1) by Lemma 2.4 and positivity, (S2) by Lemma 4.5 and (V6), (S3) by (V7) (the new deep tail consists of the old deep tail and the old coordinate $-K_b$, divided by G), (S4) by Lemma 4.2(b), since the new coordinate $k > K_a$ is $Y_{k+1}(s_*)/G \leq q_{k+1}/G_{\text{lo}} \leq \theta_k$, and (S5) by (V8), since $\varepsilon_n \Lambda^{2\alpha}/(rG) \leq \varepsilon_n \Lambda^{2\alpha_\Lambda}/(rG_{\text{lo}}) < \varepsilon_n$. $\square$

5.2. The box

Nothing in the choice of w^* and ρ needs to be rigorous. The profile w^* is the fixed point of the inviscid step map, found by iterating the map on the truncated system with a high-order Runge–Kutta method. The radii are $\rho = \delta(I - |J|/0.9)^{-1}b$, where J is the Jacobian of the step map at w^* (the program’s derivative at the centre), $\delta = 10^{-7}$, $b_k = 1$ for $k < 0$ and $b_k = \max\{0.1 w_k^*, 10^{-13}\}$ for $2 \leq k < K_a$ (10^{-11} in place of 10^{-13} for $\Lambda = 9/5$; for $\Lambda = 13/10$, whose files come from an earlier run, $b_k = \max\{10^{\lceil \log_{10} w_k^* \rceil}, 10^{-11}\}$), and $\rho_{K_a} = 10^{-20}$. The spectral radius of $|J|$ is between 0.65 and 0.85, so $|J|\rho \leq 0.9\rho - \delta b$ and the linearized step contracts the box by the factor 0.9; the computed ratios in Table 3 are close to that value. Table 2 shows the profile near the front for $\Lambda = 13/10$.

Table 2: Part of the profile w^* and of the radii ρ for $\Lambda = 13/10$ (rounded; the files hold 17 digits). The deep part decreases slowly to $w_{-60}^* = 0.1682$, with $\rho_{-60} = 1.75 \cdot 10^{-3}$.

k	−3	−2	−1	2	3	4	5	6
w_k^*	0.74412	0.83400	0.99298	$6.179 \cdot 10^{-2}$	$5.174 \cdot 10^{-4}$	$2.081 \cdot 10^{-8}$	$1.971 \cdot 10^{-17}$	0
ρ_k	$1.3 \cdot 10^{-6}$	$5.7 \cdot 10^{-7}$	$2.5 \cdot 10^{-7}$	$1.5 \cdot 10^{-8}$	$2.5 \cdot 10^{-10}$	$2.5 \cdot 10^{-14}$	$1.0 \cdot 10^{-18}$	10^{-20}

5.3. Results

All five cases pass. Table 3 lists the main intermediate quantities. The truncation error at the front is tiny ($z_1 \leq 10^{-60}$ for $\Lambda = 13/10$), because the error at the deep end is damped over many shells before it reaches the front, and the error at the ahead end is of size θ . The largest perturbation is at the deep end of the window.

Table 3: Data of the verification (rounded up). $\sup \tilde{s}$: upper end of the enclosure of the crossing time; λ : growth rate in Lemma 4.3; z_{-K_b} : perturbation bound at the deep end; worst ratio: largest value of $|\text{image} - w_k^*|/\rho_k$ in (V6); ε -factor: the left side of (V8); time: wall-clock seconds on one core.

Λ	$\sup \tilde{s}$	s_{end}	λ	z_{-K_b}	ξ/m	worst ratio	ε -factor	time
6/5	0.5517897	0.5717897	4.48	$1.6 \cdot 10^{-4}$	$2.8 \cdot 10^{-82}$	0.9000	0.99977	7.9
13/10	0.4996513	0.5196513	4.74	$6.3 \cdot 10^{-5}$	$2.3 \cdot 10^{-60}$	0.8999	0.99971	4.3
3/2	0.4173915	0.4673915	5.71	$3.2 \cdot 10^{-6}$	$2.9 \cdot 10^{-95}$	0.8996	0.99960	3.1
17/10	0.3552689	0.3752689	6.36	$1.1 \cdot 10^{-6}$	$3.1 \cdot 10^{-94}$	0.9542	0.99970	1.9
9/5	0.3295387	0.3495387	6.73	$2.2 \cdot 10^{-7}$	$6.0 \cdot 10^{-91}$	0.8994	0.99922	2.0

5.4. Implementation notes

The program uses CAPD 6.1.0 (commit 2f06098) with interval arithmetic in double precision. The vector field is written with CAPD’s automatic differentiation, with R_k and E_k as interval parameters. Two details matter for checking the code.

First, CAPD’s high-order enclosure guesses a trial box for the remainder of each Taylor step and then validates it; the guess adds 10^{-300} to the estimated remainder. For windows with more than about thirty shells, the deep coordinates have remainders below that size and the guess fails repeatedly, so the step size collapses. Our copy of the enclosure routine (cap/MyHOE.h) adds 10^{-30} instead. This only enlarges the trial box; the inclusion test that validates it is unchanged, so the enclosures remain rigorous.

Second, the matrix M and the vector ζ in (V2) are computed in floating point (the vector by a Gauss–Seidel iteration for $(\lambda_0 I - M)\zeta = \bar{e}$), but M is taken from upper bounds of the interval Jacobian, and λ and η are then computed from M and ζ in interval arithmetic, so the hypotheses of Lemma 4.3 hold exactly.

5.5. Independent checks

The scalar inequalities (V1), (V4)–(V6), (V8), the second inequality of (V7) and the condition $h < 0$ on $\bar{B}$ in (V3) were recomputed from the logged enclosures in 60-digit interval arithmetic by a separate Python program, with the profile and radii read from the input files rather than from the verifier’s output. A separate C program, in long double precision and with a fixed-step Runge–Kutta method, iterated the full model (all deep shells, fourteen ahead shells, the viscous term) for 150 steps from two corners of the box for $\Lambda = 13/10$ and $\alpha = 3/8$: the state stayed within 0.891 of the radii, and G stayed in $[1.0240158, 1.0240160]$; a Python program with an adaptive method gave the same numbers. A Julia program computed $\kappa(\Lambda)$ independently and agrees with Table 4 to twelve digits. Details are in Appendix A.

6. Numerical observations

The statements in this section are not proved; they come from the same fixed-point computation as the profile w^* , run for many ratios.

Table 4: Growth factor $\kappa(\Lambda)$ of the inviscid front (non-rigorous) and the exponent $\alpha^*(\Lambda) = 1/3 + \log \kappa / (2 \log \Lambda)$ from (1.5).

Λ	$\kappa(\Lambda)$	α^*	γ	Λ	$\kappa(\Lambda)$	α^*	γ
1.10	1.003483	0.3516	0.2968	1.60	1.022497	0.3570	0.2860
1.20	1.016285	0.3776	0.2447	1.70	1.015997	0.3483	0.3034
1.25	1.020764	0.3794	0.2412	1.80	1.007477	0.3397	0.3207
1.30	1.024016	0.3786	0.2429	1.85	1.002607	0.3355	0.3291
1.40	1.027165	0.3732	0.2537	1.87	1.000562	0.3338	0.3324
1.45	1.027238	0.3695	0.2610	1.88	0.999520	0.3330	0.3341
1.50	1.026428	0.3655	0.2690	2.00	0.986146	0.3233	0.3535

The factor $\kappa(\Lambda)$ increases from 1 near $\Lambda = 1$ to a maximum 1.0273 near $\Lambda = 1.42$, decreases, and crosses 1 at $\Lambda_c \approx 1.8754$ (linear interpolation between 1.87 and 1.88). For $\Lambda > \Lambda_c$ the fixed point has $\kappa < 1$; a front from generic data then does not follow it, and direct simulations of the inviscid model for $\Lambda = 1.9, 2.2, 2.5, 3$ and 4 show fronts with $\kappa = 1$ to six digits, that is, Kolmogorov scaling. Direct simulations of the viscous model for $\Lambda = 13/10$, started from the front profile, show growth of $\Lambda^{(1-2\alpha)n} u_n$ over 900 shells for $\alpha = 0.36, 0.37, 0.375$ and decay for $\alpha = 0.385$, consistent with $\alpha^*(13/10) = 0.3786$.

Conjecture 6.1. *For $1 < \Lambda < \Lambda_c$, the solutions with data near the front profile lose regularity in finite time if and only if $\alpha < \alpha^*(\Lambda) = 1/3 + \log \kappa(\Lambda) / (2 \log \Lambda)$; for $\Lambda \geq \Lambda_c$ the corresponding threshold is $1/3$.*

Theorem 1.1 proves the “if” direction at five ratios, up to the small gap between α_Λ and $\alpha^*(\Lambda)$ visible in Table 1. The “only if” direction would require a regularity argument for α just above $\alpha^*(\Lambda)$, which is far beyond (1.2): for $\Lambda = 13/10$ the known regularity

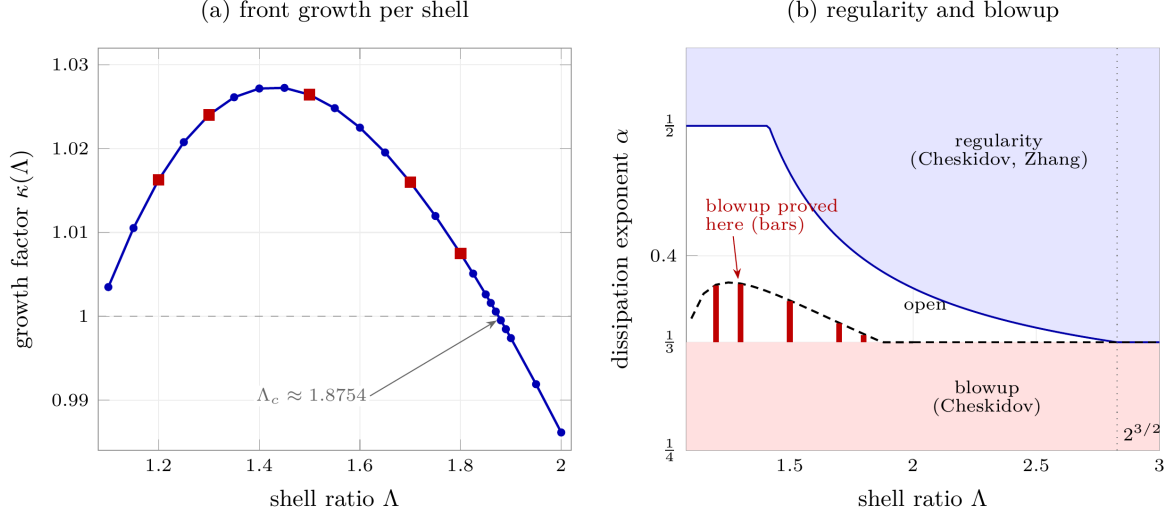


Figure 2: (a) The growth factor $\kappa(\Lambda)$ of the self-similar front (Table 4); squares mark the five verified ratios. (b) Known results in the (Λ, α) plane: blowup for $\alpha < 1/3$ [10]; regularity above $\min\{1/2, \max\{1/3, 1/4 + \log 2/(8 \log \Lambda)\}\}$ [10, 9]; the bars are the ranges $[1/3, \alpha_\Lambda]$ proved here, and the dashed curve is $\max\{1/3, \alpha^*(\Lambda)\}$.

starts at $\alpha = 1/2$. Other data might blow up for larger α through a different mechanism; we have no evidence either way.

7. Open questions

- (1) Prove Theorem 1.1 for every Λ in an interval, for instance by treating Λ as an interval parameter. The verification is fast enough for a covering of $[1.1, 1.85]$ by small intervals, but the window sizes K_b, K_a must grow as $\Lambda \rightarrow 1$ and the margin $\kappa - 1$ vanishes as $\Lambda \rightarrow \Lambda_c$.
- (2) Determine the threshold for $\Lambda \in [\Lambda_c, 2^{3/2}]$, where Conjecture 6.1 predicts $1/3$ and (1.2) gives regularity only above $1/4 + \log 2/(8 \log \Lambda)$.
- (3) Prove regularity for some $\alpha < 1/2$ when Λ is small, or find blowup above $\alpha^*(\Lambda)$.
- (4) Explain the bifurcation at Λ_c , where the anomalous front meets the Kolmogorov front, in the language of the renormalization picture of [17, 18, 19].

Data availability

All code, input files and output logs are openly available in the repository and its Zenodo archive [12].

Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) in order to write and run the programs, to write the Lean proofs and to draft the text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Appendix A. Code, data and checks

The repository <https://github.com/creelie/eNvrs>, archived on Zenodo at [doi:10.5281/zenodo.23247974](https://doi.org/10.5281/zenodo.23247974) [12] contains:

- `cap/verify_step.cpp`, `cap/MyHOE.h`: the verifier; `cap/run_all.sh` builds it and runs the five cases; `cap/cases/`: profiles and radii; `cap/logs/`: the output, with all enclosures; `scripts/build_capd.sh` builds the CAPD version used.
- `crosscheck/recheck.py`: the interval re-check of the scalar inequalities; `crosscheck/front_steps.c`: the C iteration of the full model; `crosscheck/kappa.jl`: the Julia computation of $\kappa(\Lambda)$; `numerics/`: the Python programs for the fixed point, Table 4 and the simulations of Section 6.
- `lean/DyadicNS.lean`: Lean 4 proofs, with Mathlib, of the induction in Lemma 4.2(b), the scalar case of Lemma 4.1, the supersolution inequality in the proof of Lemma 4.3, the time estimate of Lemma 4.4(b), the monotonicity of ε_n , the equivalence (1.5), the summation of the step times and the divergence of the weighted amplitudes. Each theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The interval computation is not formalized.

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